I1 — Inorganic Chemistry I

1. Periodicity

1.1 Classification of the Periodic Table

The periodic table arranges elements in order of increasing atomic number. **Periods** are horizontal rows (Period 1 has 2 elements, Periods 2 and 3 have 8 each, Period 4 has 18). **Groups** are vertical columns. Elements in the same group have the same number of outer-shell electrons and therefore similar chemical properties.

The periodic table is divided into blocks based on the sub-shell being filled:

- **s-block:** Groups 1 and 2 (and helium) — outer electrons in an s sub-shell. Includes the alkali metals (Group 1) and alkaline earth metals (Group 2).
- **p-block:** Groups 13 to 18 (old numbering: 3 to 0) — outer electrons in a p sub-shell. Includes non-metals, metalloids, and some metals.
- **d-block:** The transition metals — electrons filling the 3d (or 4d, 5d) sub-shell. These are found between Groups 2 and 13.
- **f-block:** The lanthanides and actinides — electrons filling f sub-shells. Usually shown separately at the bottom of the table.

1.2 Period 3 Trends

Atomic radius: Decreases across Period 3 from Na to Cl. The nuclear charge increases by +1 for each successive element (one extra proton), but the shielding remains approximately the same because electrons are added to the same shell (3rd shell). The increased effective nuclear charge pulls the outer electrons closer to the nucleus, reducing the atomic radius.

First ionisation energy: Generally increases across Period 3 from Na to Ar due to increasing nuclear charge and decreasing atomic radius with approximately constant shielding.

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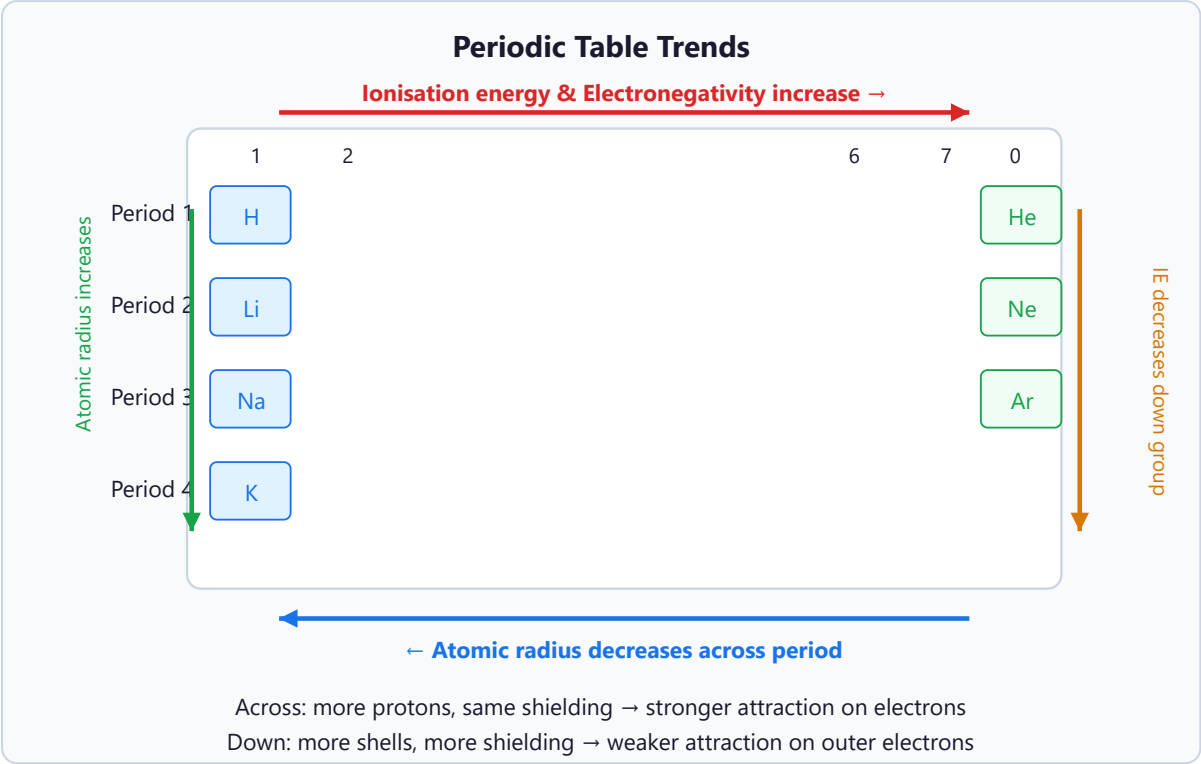
Element	Na	Mg	Al	Si	P	S	Cl	Ar
1st IE (kJ mol^{-1})	496	738	578	789	1012	1000	1251	1521
Config. (outer)	$3s^1$	$3s^2$	$3p^1$	$3p^2$	$3p^3$	$3p^4$	$3p^5$	$3p^6$

Two anomalies (drops) in the general trend across Period 3:

- **Mg (738) > Al (578):** Aluminium loses a 3p electron, which is in a higher-energy sub-shell and further from the nucleus than magnesium's 3s electron. The 3p electron is also slightly shielded by the full $3s^2$ pair, so less energy is needed to remove it. This is sometimes called the "s to p" drop.
- **P (1012) > S (1000):** Sulfur has a paired electron in one of its 3p orbitals (configuration: $3p^4$ means one orbital has a pair). The extra electron-electron repulsion between this paired pair makes one of sulfur's electrons easier to remove than one of phosphorus's three unpaired 3p electrons ($3p^3$, half-filled, more stable). This is sometimes called the "pairing" drop.

Exam Tip: When explaining IE anomalies across a period, you MUST: (1) State the specific sub-shell involved for each element (e.g. "Al loses a 3p electron, Mg loses a 3s electron"); (2) Explain WHY one is easier to remove (higher energy sub-shell / electron-electron repulsion from pairing); (3) Give the electron configurations. Generic answers about "nuclear charge" will not earn the marks.

Electronegativity: Increases across Period 3 from Na to Cl. Noble gases are excluded as they do not generally form bonds. The increase is due to increasing nuclear charge and decreasing atomic radius — atoms have a stronger pull on bonding electrons.



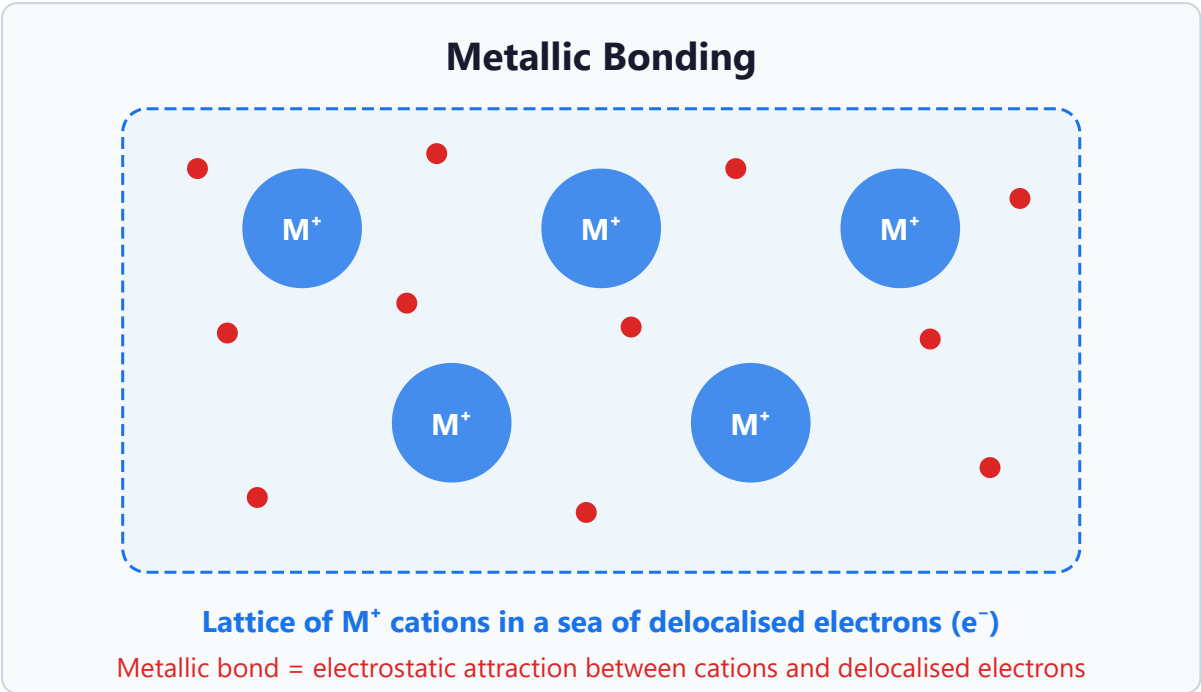
1.3 Melting Point Trend Across Period 3

Element	Na	Mg	Al	Si	P_4	S_8	Cl_2	Ar
Structure	Metallic	Metallic	Metallic	Giant covalent	Simple molecular	Simple molecular	Simple molecular	Monatomic
Melting point (K)	371	923	933	1687	317	392	172	84

Explanation of the melting point trend:

- **Na to Al (metals, increasing):** Metallic bonding becomes stronger because: (a) the charge on the metal cation increases ($Na^+ \rightarrow Mg^{2+} \rightarrow Al^{3+}$), (b) the number of delocalised electrons per atom increases (1 → 2 → 3), (c) the ionic radius decreases. All three factors strengthen the electrostatic attraction between cations and the sea of delocalised electrons, requiring more energy to break.
- **Si (very high):** Silicon has a giant covalent (macromolecular) structure similar to diamond. Each Si atom is covalently bonded to four others in a tetrahedral arrangement. Many strong Si-Si covalent bonds must be broken to melt it, requiring a great deal of energy.

- **P_4 to Ar (low, decreasing):** These are simple molecular or monatomic substances. Only weak London (dispersion) forces between molecules need to be overcome to melt them — the strong covalent bonds within molecules do NOT break. S_8 has a slightly higher melting point than P_4 because it has more electrons and greater surface area, giving stronger London forces. Cl_2 is smaller with fewer electrons, and Ar (monatomic, fewest electrons of all) has the lowest melting point.



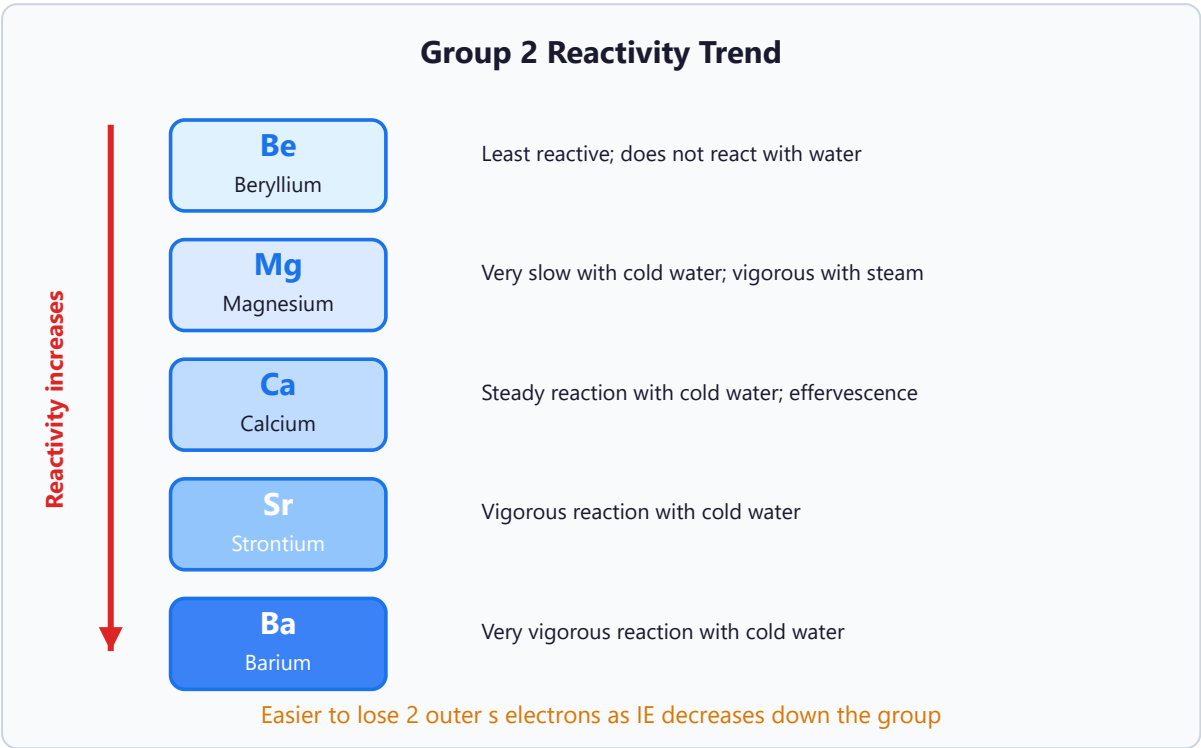
Exam Tip: When comparing melting points across a period, always state the type of structure (metallic / giant covalent / simple molecular), the type of bonding or forces being broken, and explain why it needs more or less energy. Never say "covalent bonds break" for simple molecular substances — it is the intermolecular forces (London forces) that are overcome on melting.

2. Group 2 — The Alkaline Earth Metals

2.1 Trends Down Group 2

Property	Trend down the group	Explanation

Atomic radius	Increases	More electron shells added, so outer electrons are further from the nucleus.
First ionisation energy	Decreases	Outer electrons are further from the nucleus and more shielded by additional inner shells. Easier to remove despite increased nuclear charge — the shielding and distance effects dominate.
Electronegativity	Decreases	Bonding electrons are further from the nucleus and more shielded, so atoms attract them less strongly.
Reactivity	Increases	It becomes easier to lose the two outer s electrons as IE decreases. Atoms are more readily oxidised to form M^{2+} ions.
Melting point	Generally decreases (with some irregularities)	Larger cations with the same charge — metallic bonding becomes weaker as the charge density of the cation decreases.



2.2 Reactions of Group 2 Elements

With water: Group 2 metals react with water to form a metal hydroxide and hydrogen gas. Reactivity increases down the group as ionisation energies decrease.

$$M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq) + H_2(g)$$

- **Beryllium:** Does not react with water or steam. It has a protective oxide layer and very high ionisation energies.
- **Magnesium:** Reacts very slowly with cold water (thin layer of $Mg(OH)_2$ forms, a few bubbles of H_2). Reacts vigorously with steam: $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$. The magnesium burns with a bright white light.
- **Calcium:** Reacts steadily with cold water, producing bubbles of hydrogen and a white precipitate/milky solution of $Ca(OH)_2$ (sparingly soluble).
- **Strontium:** Reacts more vigorously than calcium with cold water.
- **Barium:** Reacts vigorously with cold water, producing $Ba(OH)_2$ (more soluble, so the solution is clearer) and hydrogen gas.

With oxygen: All Group 2 metals burn in air to form white solid metal oxides:

$$2M(s) + O_2(g) \rightarrow 2MO(s)$$

Magnesium burns with a brilliant white flame — it is used in flares and fireworks.

With dilute acids: Group 2 metals react with dilute acids to form a salt and hydrogen. For example:

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

2.3 Solubility Trends of Group 2 Compounds

Compound type	Solubility trend down Group 2	Key examples
Hydroxides $M(OH)_2$	**Increases**	$Mg(OH)_2$ nearly insoluble; $Ca(OH)_2$ slightly soluble; $Ba(OH)_2$ soluble

Sulfates MSO_4	**Decreases**	$MgSO_4$ very soluble (Epsom salts); $CaSO_4$ slightly soluble (gypsum); $BaSO_4$ very insoluble

Real-world application: Barium sulfate is so insoluble that it is safely used as a "barium meal" in medical X-ray imaging of the digestive system. Despite Ba^{2+} ions being toxic, $BaSO_4$ passes through the gut without dissolving, so it cannot release harmful ions into the bloodstream. The dense barium atoms absorb X-rays strongly, making the gut visible on the image.

Exam Tip: The solubility trends go in opposite directions: hydroxides become MORE soluble down the group, sulfates become LESS soluble. A common exam question asks you to explain why barium sulfate is used in medicine — you must state that it is insoluble, so Ba^{2+} ions are not released.

2.4 Thermal Decomposition of Group 2 Carbonates and Nitrates

Carbonates: Group 2 carbonates decompose on heating to form the metal oxide and carbon dioxide:

$$MCO_3(s) \rightarrow MO(s) + CO_2(g)$$

Thermal stability **increases** down the group — higher temperatures are needed to decompose them. $BeCO_3$ decomposes most easily; $BaCO_3$ requires the highest temperature.

Explanation using polarisation: The smaller, more highly charged cations near the top of the group (like Be^{2+} and Mg^{2+}) have a higher charge density. They polarise (distort) the electron cloud of the large CO_3^{2-} anion more strongly, weakening the C-O bonds within the carbonate and making it decompose more easily at a lower temperature. Larger cations (like Ba^{2+}) have lower charge density and polarise the carbonate less, so decomposition requires more energy.

Nitrates: Group 2 nitrates decompose on heating to form the metal oxide, nitrogen dioxide (brown gas), and oxygen:

$$2M(NO_3)_2(s) \rightarrow 2MO(s) + 4NO_2(g) + O_2(g)$$

The same trend in thermal stability applies — stability increases down the group for the same polarisation reasons.

Exam Tip: When explaining thermal stability trends, always use the concept of polarisation. State that the smaller cation has a higher charge density, which polarises the anion more, weakening its bonds and making it easier to decompose. Draw a diagram showing the cation distorting the anion's electron cloud if possible.

2.5 Flame Tests for Group 1 and 2 Metals

Metal ion	Flame colour
Li^+	Crimson (red)
Na^+	Yellow
K^+	Lilac
Ca^{2+}	Orange-red
Sr^{2+}	Red
Ba^{2+}	Green
Cu^{2+}	Blue-green

How flame tests work: The heat of the flame excites electrons in the metal ion to higher energy levels. When these electrons fall back to their ground state, they emit photons of light with specific wavelengths corresponding to the energy gap between levels. Different elements have different electronic structures and therefore different energy gaps, producing characteristic colours.

Method: Dip a clean nichrome wire loop in concentrated HCl (to clean it), then dip it in the solid sample, and hold it in a blue Bunsen flame. Observe the colour.

2.6 Uses of Group 2 Compounds

- **$Mg(OH)_2$ (milk of magnesia):** Antacid for neutralising excess stomach acid. $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$.

- **$Ca(OH)_2$ (slaked lime):** Neutralising acidic soils in agriculture (liming). Also used to treat acidic industrial waste water and in flue gas desulfurisation to remove SO_2 from power station emissions: $Ca(OH)_2 + SO_2 \rightarrow CaSO_3 + H_2O$.
- **CaO (quicklime):** Used in cement and concrete manufacture, and as a drying agent.
- **$CaCO_3$ (limestone):** Used in the blast furnace to remove acidic SiO_2 impurities from iron ore. $CaCO_3$ decomposes to CaO, which reacts with SiO_2 to form slag ($CaSiO_3$).
- **$BaSO_4$:** Barium meal for X-ray imaging of the digestive tract.

3. Group 7 — The Halogens

3.1 Physical Properties and Trends

Halogen	State at RTP	Colour	Electron config (outer)	Electronegativity
F_2	Pale yellow gas	Pale yellow	$2s^2\ 2p^5$	4.0
Cl_2	Yellow-green gas	Yellow-green	$3s^2\ 3p^5$	3.0
Br_2	Red-brown liquid	Red-brown	$4s^2\ 4p^5$	2.8
I_2	Dark grey solid	Purple vapour	$5s^2\ 5p^5$	2.5

Boiling points increase down the group because the molecules become larger with more electrons, so London (dispersion) forces become stronger and require more energy to overcome. This explains the trend from gas (F_2, Cl_2) to liquid (Br_2) to solid (I_2).

3.2 Trends in Electronegativity and Reactivity

Electronegativity decreases down Group 7. Fluorine is the most electronegative element on the periodic table. Going down, atoms have larger atomic radius and more inner-shell shielding, so they attract bonding electrons less strongly.

Reactivity decreases down the group. Halogens react by gaining one electron to form halide ions (X^-). Their oxidising ability (ability to gain electrons) decreases down the group because:

- Atomic radius increases — the incoming electron enters a shell further from the nucleus.
- Shielding increases — inner shells shield the incoming electron from the nuclear charge.
- The attraction between the nucleus and the incoming electron weakens.

Oxidising agent: A substance that causes another substance to be oxidised, by gaining electrons itself (being reduced). Halogens are powerful oxidising agents. Oxidising ability: $F_2 > Cl_2 > Br_2 > I_2$.

3.3 Displacement Reactions of Halogens

A more reactive halogen will displace a less reactive halide ion from an aqueous solution of its salt. This demonstrates the relative oxidising power of the halogens — the stronger oxidising agent oxidises the weaker halide.

Halogen added	KCl(aq)	KBr(aq)	KI(aq)
Cl_2(aq)	No reaction	Orange/brown colour (Br_2 displaced)	Brown colour (I_2 displaced)
Br_2(aq)	No reaction	No reaction	Brown/black colour (I_2 displaced)
I_2(aq)	No reaction	No reaction	No reaction

Ionic equations for displacement reactions:

$$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$$

$$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$$

$$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$$

In each case, the halogen is the oxidising agent (it is reduced: $Cl_2 + 2e^- \rightarrow 2Cl^-$), and the halide ion is the reducing agent (it is oxidised: $2Br^- \rightarrow Br_2 + 2e^-$).

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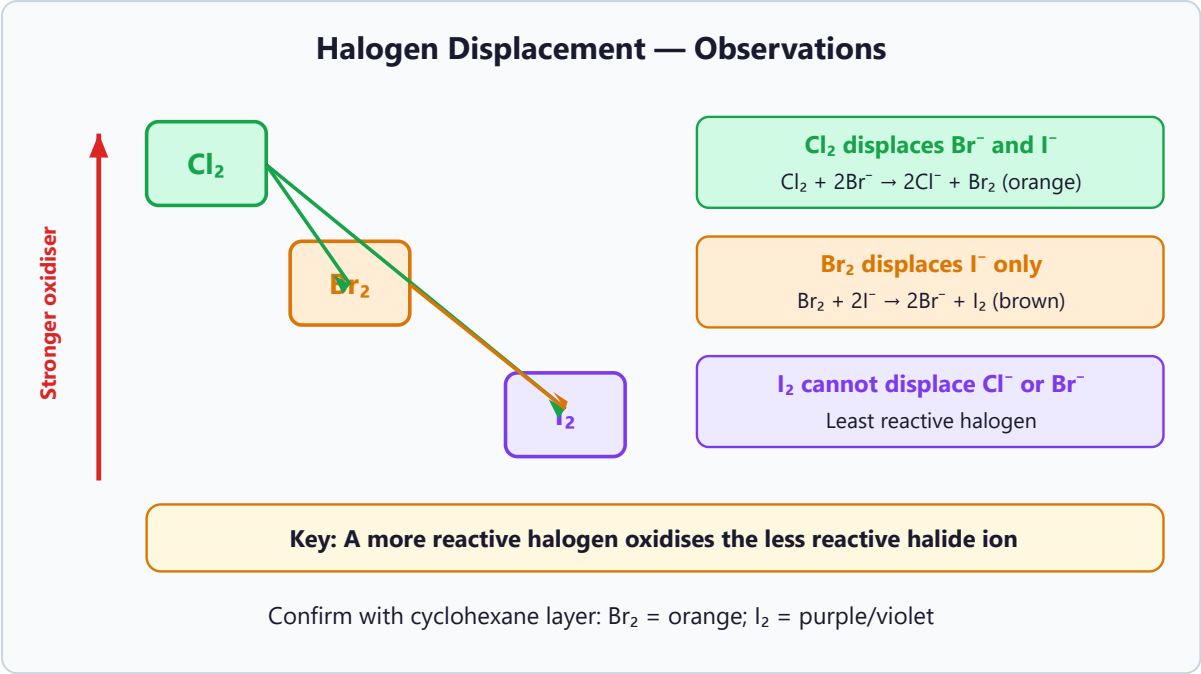

Exam Tip: In displacement questions, always write the ionic equation and identify the oxidising and reducing agents. State which species is oxidised and which is reduced. Adding a non-polar solvent (cyclohexane) helps confirm the identity of the displaced halogen — Br_2 gives an orange organic layer, I_2 gives a purple/violet organic layer.

3.4 Halide Ions as Reducing Agents

Halide ions can act as reducing agents by donating electrons. Reducing ability **increases** down the group: $I^- > Br^- > Cl^- > F^-$. This is because the outer electron is further from the nucleus and more easily lost (the ion is larger, and the electron being removed is more shielded).

Reactions of solid sodium halides with concentrated sulfuric acid:

Halide	Initial reaction	Redox?	Further products	Observations
NaCl	$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$	No redox (acid-base only)	HCl gas only	Steamy white fumes
NaBr	$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$	Yes — HBr reduces H_2SO_4	Br_2, SO_2, H_2O	Steamy fumes + orange/brown fumes (Br_2) + choking gas (SO_2)

NaI	$NaI + H_2SO_4 \rightarrow NaHSO_4 + HI$	Yes — HI reduces further	I_2, SO_2, S, H_2S	Purple/black solid (I_2), yellow solid (S), rotten egg smell (H_2S)

Explanation:

- **NaCl:** Cl^- is too weak a reducing agent to reduce H_2SO_4. Only an acid-base reaction occurs, producing HCl gas (steamy fumes).
- **NaBr:** Br^- is a stronger reducing agent. After the initial acid-base reaction produces HBr, the HBr is strong enough to reduce the sulfuric acid: $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$. Sulfur goes from +6 to +4 in SO_2.
- **NaI:** I^- is the strongest reducing agent in the group. HI reduces H_2SO_4 even further — sulfur goes from +6 to +4 (SO_2), then to 0 (S), and finally to -2 (H_2S). Multiple products are observed: purple I_2, yellow S, and H_2S with its distinctive rotten egg smell.

3.5 Tests for Halide Ions (Silver Nitrate Test)

Method: Add dilute nitric acid (HNO_3) to the solution first (to remove interfering ions like carbonate and hydroxide that would form precipitates with silver), then add a few drops of silver nitrate solution ($AgNO_3$).

Halide ion	Silver halide precipitate	Colour	Solubility in dilute NH_3	Solubility in conc. NH_3
Cl^-	AgCl	White	Soluble (dissolves)	Soluble
Br^-	AgBr	Cream (off-white)	Insoluble	Soluble (dissolves)
I^-	AgI	Yellow	Insoluble	Insoluble

$$Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$$

The ammonia test helps distinguish the precipitates if the colours are ambiguous. AgCl dissolves in dilute NH_3 because the ammonia forms a soluble complex: $AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-$. AgBr requires concentrated NH_3, and AgI does not dissolve at all.

Exam Tip: You MUST state that dilute nitric acid is added first — NOT hydrochloric acid (which would add chloride ions and give a false positive). NOT sulfuric acid either (which could precipitate insoluble sulfates). State the colour AND the formula of the precipitate for full marks.

3.6 Disproportionation Reactions of Chlorine

Disproportionation: A reaction in which the same element in a single species is simultaneously oxidised and reduced, forming products in which that element has both a higher and a lower oxidation state.

Chlorine with water:

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$$

Chlorine (oxidation state 0) is reduced to -1 in HCl and oxidised to +1 in HClO (hypochlorous acid). This is a disproportionation because the same element (Cl) is both oxidised and reduced in the same reaction. HClO is a powerful oxidising agent that kills bacteria.

Chlorine with cold dilute NaOH:

$$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$

This produces sodium chlorate(I) (NaClO), the active ingredient in household bleach and swimming pool disinfectant. Chlorine disproportionates from 0 to -1 (in NaCl) and +1 (in NaClO).

Chlorine with hot concentrated NaOH:

$$3Cl_2(aq) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$

At higher temperature, chlorate(V) ions (ClO_3^-) are formed instead, with Cl in oxidation state +5.

3.7 Uses of Chlorine and the Ethical Considerations

Water treatment and purification: Chlorine is added to drinking water supplies to kill bacteria and other pathogens. The chlorine reacts with water to form HClO, which is the actual bactericidal agent. This

prevents waterborne diseases such as cholera, typhoid, and dysentery.

Benefits vs risks (an ethical consideration often tested in exams):

- **Benefits:** Chlorination has virtually eliminated waterborne diseases in developed countries, saving millions of lives. It is cheap, effective, and provides residual protection as water travels through pipes.
- **Risks:** Chlorine can react with organic compounds in the water to form small amounts of chlorinated hydrocarbons (such as trihalomethanes, THMs), some of which may be carcinogenic with prolonged exposure.
- **Conclusion:** The overwhelming health benefits of chlorinating water (preventing deadly epidemics) far outweigh the very small risk from trace chlorinated by-products. The scientific consensus supports continued chlorination.

Other uses of chlorine: Manufacturing PVC (polyvinyl chloride), solvents, pesticides, and pharmaceuticals. Sodium hypochlorite (NaClO) is used as bleach for disinfection and whitening.

> **Exam Tip:** For questions about water chlorination, present both sides of the argument (benefits and risks) but conclude that the benefits outweigh the risks. Use specific examples: name cholera/typhoid as diseases prevented, and name THMs/chlorinated hydrocarbons as the potential risk.

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